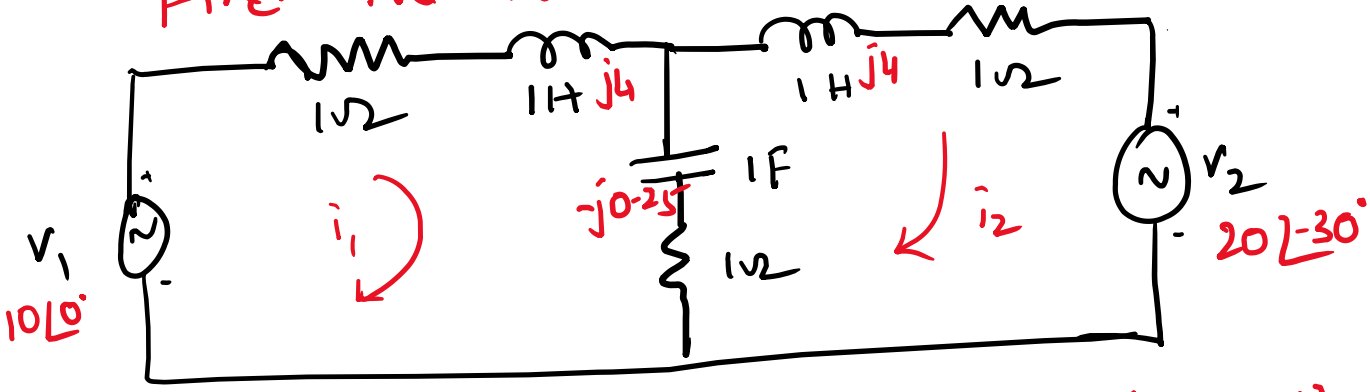
Mesh Analysis in AC

Find the mesh currents i_1, i_2



$v_1 = 10\cos 4t$ $\qquad$ $v_2 = 20\cos(4t - 30^\circ)$

Mesh 1:

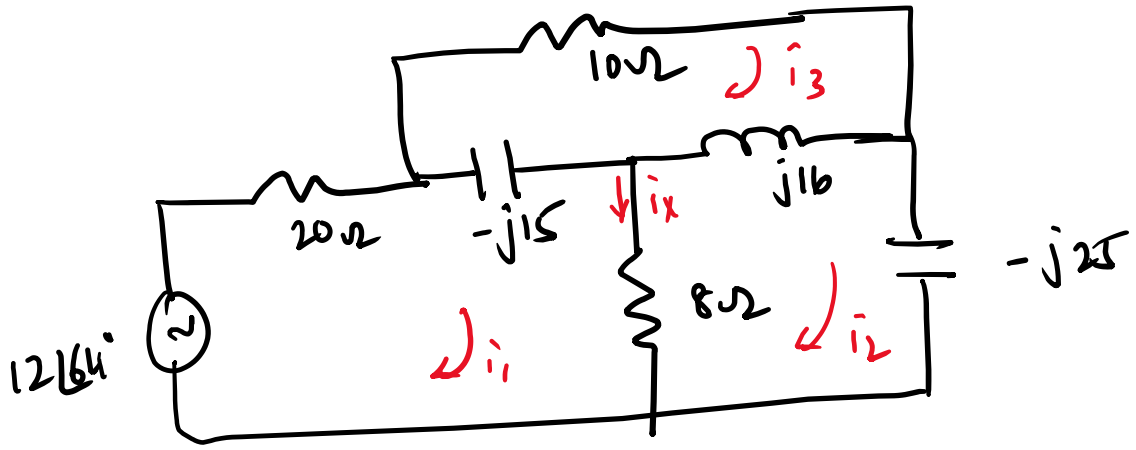
$$10\angle 0^\circ - i_1(1+j4) - (i_1 - i_2)(1 - j0.25) = 0$$

$$10 = i_1(2 + j3.75) - i_2(1 - j0.25)$$

Mesh 2: $\quad -20\angle -30^\circ - (i_2 - i_1)(1 - j0.25) - i_2(1 + j4) = 0$

$$-20\angle -30^\circ = -i_1(1 - j0.25) + i_2(2 + j3.75)$$

$$-17.32 + j10$$



Find i_1, i_2, i_3, i_x using mesh analysis.

Mesh 1: $12\angle 64^\circ - i_1(20) - (i_1 - i_3)(-j15) - (i_1 - i_2)(8) = 0$

$$12\angle 64^\circ = i_1(28 - j15) - 8i_2 + j15\,i_3$$

Mesh 2: $-(i_2 - i_1)\,8 - (i_2 - i_3)(j16) - i_2(-j25) = 0$

$$8i_1 - i_2(8 - j9) + j16\,i_3 = 0$$

Mesh 3: $-10i_3 - (i_3 - i_2)(j16) - (i_3 - i_1)(-j15) = 0$.

$$j15\,i_1 - j16\,i_2 + i_3(10 + j) = 0$$

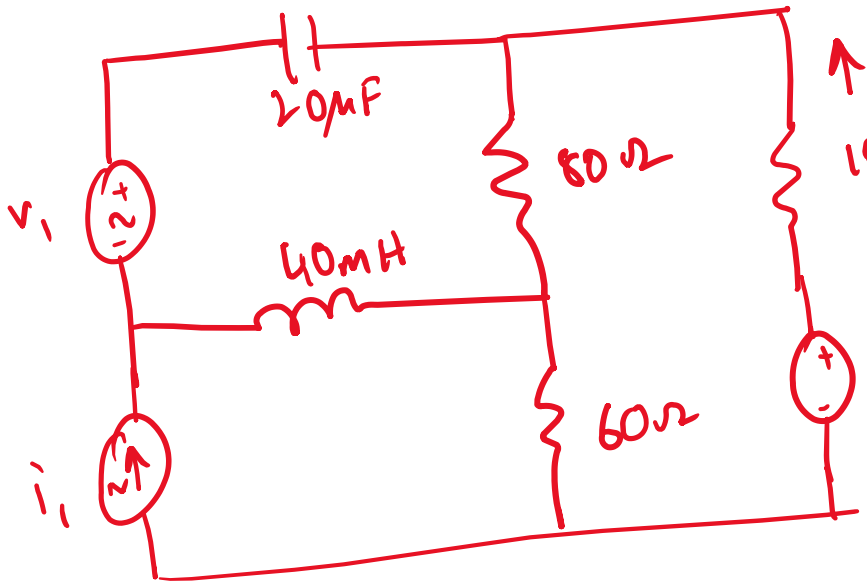
$i_1 = 381.4\angle 109.6^\circ$ mA

$i_2 = 344.3\angle 124.4^\circ$ mA

$i_3 = 145.5\angle -60.42^\circ$ mA

$i_x = i_1 - i_2$

$= 100.5\angle 48.5^\circ$ mA



$v_1 = 50\cos 2000t$ V

$i_1 = 2\sin 4000t$ A
$= 2\cos(4000t - \frac{\pi}{2})$

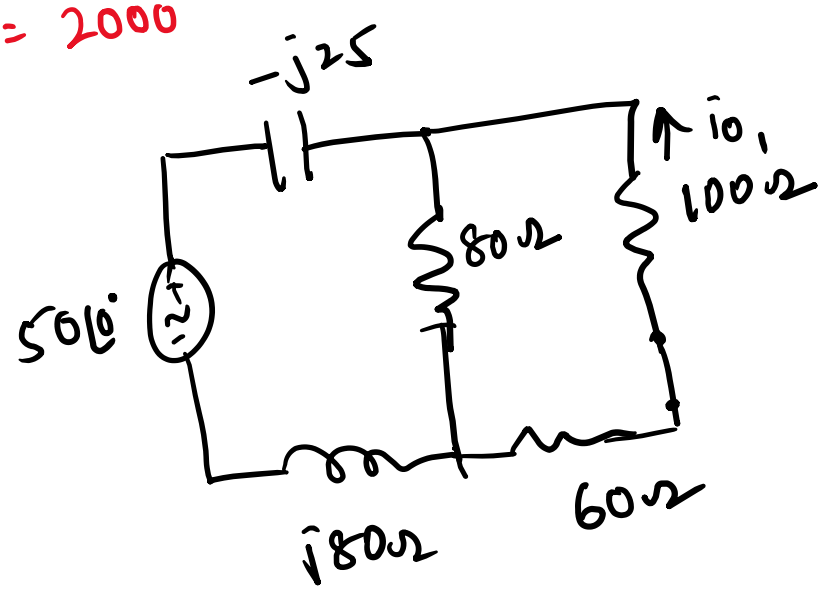
Find i_o using superposition.

$i_o = i_{o1} + i_{o2} + i_{o3}$

V_1 is ON only. $\omega = 2000$

40mH → $j80$

20mF → $-j25$

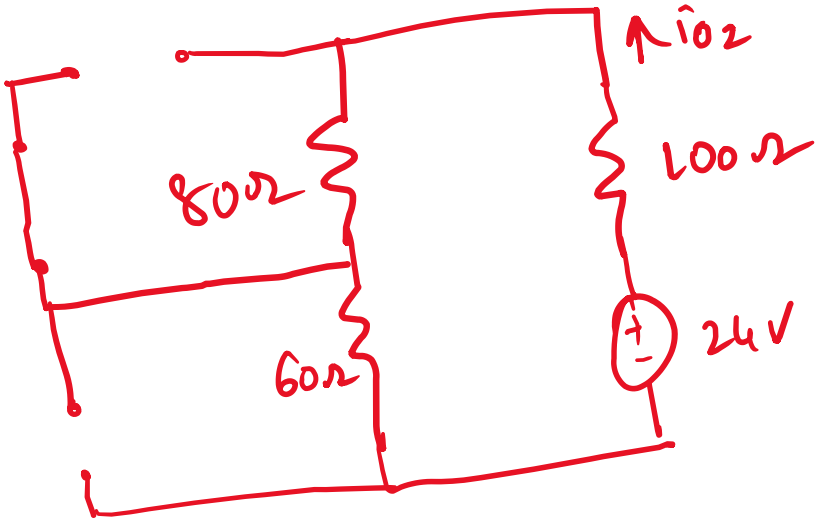


V_{DC} ON :

At DC

Cap → open

Ind → short.



$i_{o2} = \frac{24}{100 + 80 \,\|\, 60}$

i_{AC} on: $\omega = 4000$

40mH → $j160$

20μF → $-j12.5$

$2\angle -90^\circ$ A

$\hat{i}_{o3} = -\hat{i}_3$

$\hat{i}_2 = 2\angle -90^\circ$

